Mini Project Part A - Antimicrobial Resistance

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1 Introduction

The aim of this appraisal is to highlight the global challenge of antibiotic resistance and shed light on its impact on public health. In addition we will explore the multifaceted background of drug resistance and the complex drivers for its expansion. Albeit, the revolutionary breakthrough of antibiotics now poses a serious global threat due to the resulting evolution of antibiotic resistant strains(1).

The proliferation of inordinate antibiotic usage since the 1940s has proven to be consequential, as antibiotic resistance (AR) is at present a global health problem(2). Recent studies illustrate that new resistance mechanisms are evolving rapidly, and former antibiotic therapy methods that were favourable for eradicating certain strains are now ineffective. As a case in point, various strains of S. aureus show the ability to easily acquire drug resistance with the exception of vancomycin. In 1960, the first case of MRSA was detected; subsequently, S. aureus is now the leading potent opportunistic pathogen. This denotes that the emergence of variants incurable to existing antibiotics is on the rise. Additionally, global surveillance shows that infections induced by non-susceptible bacteria accelerate at up to two-fold higher rates in comparison to similar infections generated by susceptible bacteria(3). This underscores the critical need to understand the complexity of evolving bacteria, as epidemiological studies show that clinical isolates are increasingly multiply drug-resistant. Henceforth, it is imperative to practice appropriate antibiotic therapy, as comprehensive studies indicate the rising aftermath of drug resistance, from decreased therapeutic effectiveness and inappropriate empirical therapy to economic burdens.

2 Appraisal

2.1 Disparities across different regions

The disparity in the prevalence of antibiotic resistance globally is largely due to geographical differences. Observational studies show that lower-income regions are disproportionately affected, despite this, there is a paucity of research on these issues(4). This presents a concern as developing countries are behind in the fight against antimicrobial resistance. A pivotal study in a hospital in Fiji showcased the high rates of Staphylococcus aureus bacteremia (SAB), with an in-hospital mortality rate of over 25%(5). It was concluded that MRSA was responsible for 8.4% of nosocomial infections, highlighting the surge of resistant strains. Furthermore, a comprehensive study was conducted in India at a hospital to map the challenges they faced in implementing an antimicrobial stewardship initiative, and several outcomes surfaced that shed light on their elevated usage of antibiotics(6). Two key rationales for the improper distribution of antibiotics were patient satisfaction and non-compliant prolongation of treatment.It was reported that it was not unusual for patients to demand antibiotics despite it not being necessary. Comparisons between these prospective studies establish that education is fundamental to tackle this phenomenon(7). Thus, it is plausible that AR is a global issue and no one is immune to its consequences.

2.2 Aetiology

The aetiology of antibiotic resistance is multifaceted with the main drivers in the emergence of drug resistant pathogens being the misuse and substantial usage of antibiotics. A meta-analysis study displayed that irrespective of the education level and socio-ecnomic status of a country it's a prevalent issue(8). Complacency regarding patient education and healthcare provider communication are significant factors. The self administration of antibiotics is a notable dilemma with a staggering population of 75% in low-middle income countries correlated with the figure of 66% in some USA states(9)(10). This poses a challenge as informed guidance from a healthcare professional is essential in receiving optimal outcomes from treatments. It is noted that a substantial volume of patients truncate or delay antibiotic treatment duration and alter prescribed dosages. A representative example from a report is that a ⅓ of patients do not abide by their recommended treatment plan, a ½ pause treatment when feeling better and a ⅓ store remaining antibiotics for future use. This bears considerable implications as there is an increased probability of resistant bacteria surviving and evolving leading to recurrent infections and more severe health conditions. This is substantiated by the record of 2 million patients per year experiencing nosocomial infections with a 99,000 mortality rate. The broader significance of this analysis lies in its relevance to preventing infections in hospitals by tackling misuse of antibiotics(11).

Researchers at the CDC estimated that 50 million outpatient prescriptions are unnecessary consequently contributing to the surge of antibiotic resistance(12). Aforementioned in 2.1 , many healthcare professionals act against their best interest when patients demand antibiotics even in the event when they have a viral or fungal infection(7). A prevailing problem is that many patients associate the administration of antibiotics with perceived satisfaction without any knowledge of the risks involved. This is particularly evident in low - middle income countries due to culture. Thus, addressing these perceptions are important at correcting the false perception of antibiotics as a ‘cure all’ for patients.

In accordance with the World Health Organisation it is declared that advancements of new antibiotics are both ‘stagnant’ and ‘inadequate’ with evidence that only 12 antibiotics have been approved since 2017(13). This poses a serious global challenge as antimicrobial resistance is on the rise leading to increased mortality rates. This is attributable to the lengthy pathway to approval for new projects as well as limited financial incentives and lack of funding. As well as that the timeframe of new projects progressing from preclinical to clinical stages takes 10 years to achieve which is a concern as the rate of AR is outpacing the development of new projects. Therefore the importance of this issue cannot be overstated.

2.3 Impact on Public Health

A literature review established the multifactorial global threat of antibiotic resistance, from economic burden concerns to the imperilling success of new medical advancements. The efficiency of antibiotics is vital in several fields of medicine, including neonatal intensive care and chemotherapy. It is reported that the rise in MDR infections is the leading principle in the mortality and morbidity rates amongst patients. This perspective aligns with two findings, firstly from a systemic review published by the University of Texas that showed high resistance rates in infections in cancer patients with chemotherapy-related neutropenia as well as the 2019 AR report by the CDCf (Centre for Disease Control) which outlined an estimate of over 2 million infections a year and a resulting 35,900 deaths(14)(15). This further denotes the grave impact AR has on public health.

In Europe, the substantial economic burden deriving from antibiotic resistance was estimated to be at least 1.5 billion euros, with more than 900 million euros related to hospital costs. Comparatively, in the USA, the recorded outlay was $55 billion per year, with $20 billion in direct hospital costs(16). These markedly high figures substantiate that it is a public health concern, as there are implications from increased healthcare expenditure to extended hospital stays. This poses a problem as patients will have to be administered second line or potentially third line antibiotics which are considerably more expensive and have a wider array of adverse effects. In addition, drug resistance results in prolonged illness indicating longer hospital stays , this is supported by data showing that hospital durations have prolonged from 6.4 to 12.7 days, collectively at 8 million days a year(7). Arguably this is undesirable as societal consequences due to healthcare costs , loss of wages and death will have detrimental effects on patients and communities.

2.4 Impact on the Environment

The correlation between anthropogenic pollution with the incidence of antimicrobial resistance has only been explored on the effects on human health and ecosystems in recent years. Despite limited supporting data, in recent years the marine ecosystem has been highlighted as a 'global reservoir' for AR microbes(17). It is believed that antibiotics used in coastal aquaculture enter the marine ecosystem causing the widespread of resistant microbes ultimately disrupting the ecology. A staggering 80% of ingested antibiotics reach the marine environment via wastewater channels. Corroborated by this figure it becomes apparent that this is a crucial issue. To the extent that 6 antibiotics are mentioned in a Watch List by the European Commission. Furthermore , it is approximated that 90% of microbes in the sea are resistant to a minimum of one antibiotic and 20% to five. This is a problem as it will disrupt marine ecosystems and their biodiversity. Examination on the impact on marine life was studied in the marine environment in Kuwait which evidenced a high sum of 69 % of isolates resistant to one or more antibiotics(18). Notably, the E.coli strain screened showed resistance to 22/23 of the antimicrobials(19). This evidence implies the critical need for AR screening in marine ecosystems to mitigate the impact on both aquatic life and consequently human life. Moreover , evidence indicates that concentrations of active antimicrobial agents such as sulfonamides are present in water- sludge environments(20). To illustrate this point , a comprehensive study showcased that high concentrations of sulfonamides and other agents were present in pig farm wastewater. Given the data , it is evident that antimicrobial resistance is a global health challenge universally affecting everywhere.

3 Conclusion

In conclusion, antimicrobial resistance proves to be a global threat as the cumulative evidence asserts that it has damaging impacts from human health to marine ecosystems. These findings serve as a cornerstone in understanding the implications it has on public health.

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